Ananya Naik

FPGA Design Engineer

RESUME_1534

Location: Kolkata, West Bengal, India | Email: ananya.naik.resume_1534@example.com | Phone: +91-00000-00000 (placeholder)
Total Experience: 12 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 12 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: SPI, Timing Closure, ILA, VHDL, Hardware Simulation, Xilinx Vivado, FPGA, AXI4, DSP, Python, Intel Quartus, Verilog

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

Lead FPGA Design Engineer | ZenithTrail Technologies | Jan 2022 - Present

- Verified 189 protocol corner cases with repeatable simulation suites
- Improved maximum operating frequency by 38% after timing-path optimization
- Lowered end-to-end data-path latency by 27% using pipelined logic
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

FPGA Design Engineer | WillowGrid Labs | Jan 2018 - Dec 2021

- Lowered end-to-end data-path latency by 32% using pipelined logic
- Reduced FPGA resource utilization by 18% through arithmetic sharing

Associate FPGA Design Engineer | Evergrid Technologies | Jul 2014 - Dec 2017

- Verified 688 protocol corner cases with repeatable simulation suites
- Improved maximum operating frequency by 18% after timing-path optimization

PROJECTS

Camera Data Bridge | Intel Quartus, Timing Closure, SPI, Hardware Simulation

- Designed AXI streaming logic, frame buffers and clock-domain boundaries.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

Protocol Adapter Core | Python, AXI4, Timing Closure, Intel Quartus

- Developed reusable SPI and UART modules with loopback tests.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Silverlake School of Engineering, 2014

Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

Timing Closure Training (2021)